

LEARNING SHARED ACOUSTIC ATTRIBUTES ACROSS RESPIRATORY SOUND DATASETS WITH HETEROGENEOUS ANNOTATIONS

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ABSTRACT

Respiratory sound datasets provide complementary supervision, but differ in annotation granularity and prediction targets. We propose LSAA (Learning Shared Acoustic Attributes), a joint learning framework that shares acoustic supervision while preserving native tasks. LSAA learns abnormality, crackle, and wheeze from ICBHI cycles and SPRSound events, then uses task-specific readouts for four-class cycle and binary event prediction. This allows fine sound-type annotations to supervise learning even when the output is binary. We further assess attribute transfer and auxiliary supervision on HF Lung, and external generalization on KAUH. Across three seeds with official-test-based check-point selection, LSAA improves ICBHI Score by 3.39 percentage points with a 2.13-point SPRSound decrease relative to the respective single-source models. Further comparisons support retaining fine annotations under the evaluated training recipe and reveal task-dependent effects of readout design and additional supervision.

Index Terms— respiratory sounds, shared supervision, heterogeneous annotations, acoustic attributes, cross-task learning

1. INTRODUCTION

Automated recognition of crackles and wheezes can reduce the manual effort involved in respiratory sound assessment [1]. Training these classifiers requires expert annotations that are costly to obtain at scale [2]. Existing respiratory datasets [3, 4] offer an opportunity to expand supervision by reusing labeled recordings. Their annotations describe related acoustic attributes but differ in granularity and prediction targets, so their task labels are not directly interchangeable.

To reuse annotations that are not directly interchangeable, a common strategy is to map source labels to a shared prediction target, as in SPRSound’s fusion experiments and LungMix’s cross-dataset transfer [4, 5]. This enables reuse under an aligned task definition, but coarsening labels can discard distinctions needed by another task. Representing sound composition offers a more flexible way to use label information: Multi-breath and PC-MCL employ multilabel

supervision within ICBHI [6, 7]. Across datasets, DCASE combines class mappings and masked supervision while retaining dataset-specific evaluation within a shared temporal sound-event detection system [8]. These studies provide ways to align labels and retain their information, but our setting requires sharing supervision across distinct cycle-level and event-level prediction targets. For example, SPRSound’s Wheeze annotations contain information relevant to ICBHI’s four-class cycle task, although its binary event task requires only normal/abnormal predictions [4]. The available supervision is therefore richer than the required output. We aim to jointly learn from this information while preserving each dataset’s native prediction target.

To this end, we propose LSAA (Learning Shared Acoustic Attributes). A shared encoder learns acoustic representations through abnormality, crackle, and wheeze supervision from ICBHI cycles and SPRSound events. Each annotation supervises only the attributes it supports. Task-specific readouts map the shared attribute predictions to four-class cycle and binary event outputs, allowing fine supervision to guide learning while preserving different native tasks.

In this work, we demonstrate:

1. We propose LSAA, a joint learning framework that connects heterogeneous respiratory sound annotations through shared acoustic attributes in one model.
2. We develop task-specific readouts over shared abnormality, crackle, and wheeze predictions, enabling fine-grained supervision to support distinct native outputs.
3. We evaluate native-task performance on ICBHI and SPRSound, investigate attribute transfer and auxiliary supervision on HF Lung [1], and assess external generalization on KAUH [9].

2. DATASETS AND ACOUSTIC CHARACTERISTICS

2.1. Datasets and Annotations

ICBHI and SPRSound provide LSAA’s core joint-training data. ICBHI annotates respiratory cycles as Normal, Crackle (C), Wheeze (W), or Both (C and W) [3]. SPRSound annotates seven event categories, while the event task studied

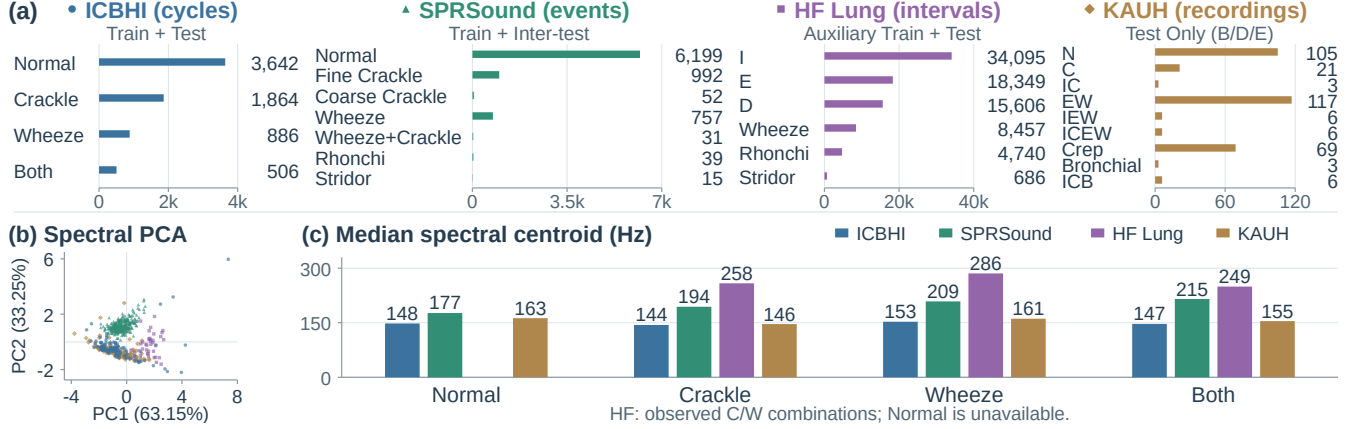


Fig. 1. Source labels and acoustic distributions. (a) Original label counts in the units and partitions shown. HF I/E/D denote inhalation, exhalation, and discontinuous adventitious sounds; KAUH retains source sound codes, and B/D/E denote Bell/Diaphragm/Extended recording versions. (b) PCA of log-transformed, jointly standardized spectral centroid, bandwidth, and peak frequency, with equal source weights. Points represent 126 ICBHI patients, 284 SPRSound train+inter patients, 39 HF source-test recording-date groups, and 112 KAUH patients. (c) Source medians by sound-label group, aggregated within recording and then patient (recording date for HF). HF source-test groups follow observed D/Wheeze annotation combinations, with no Normal group; their definition is given in Section 2.2. KAUH uses 86 patients with compatible labels; Both contains two patients. Spectral descriptors use 5-s inputs and globally centered 4-kHz analysis copies, with 80–2000 Hz Welch power (1-s Hann windows, 50% overlap).

here predicts Normal or Adventitious (abnormal) [4]. Both datasets provide patient identifiers and annotation boundaries. SPRSound’s fine C/W labels retain sound-type information relevant to ICBHI’s four-class task, even though its own prediction target is binary. This motivates sharing fine acoustic supervision while retaining each task’s prediction unit and target.

HF Lung provides timed breath-phase and adventitious-sound annotations [1]. We use its source-training partition for a separate auxiliary C/W supervision condition and its source-test partition for attribute evaluation. KAUH provides recording-level sound categories and Bell, Diaphragm, and Extended filter versions of each patient’s recording [9]. We use these recordings for external evaluation, grouping the filter versions within each patient. Figure 1(a) summarizes the annotation units, prediction tasks, and study roles.

2.2. Acoustic Characteristics

To compare acoustic inputs across sources, we prepare mono 16-kHz, 5-s inputs: ICBHI cycles, SPRSound events, and KAUH recordings are front-truncated or repeated with boundary fading, while HF recordings supply three consecutive windows. From globally centered 4-kHz analysis copies, we estimate 80–2000 Hz Welch spectra [10] and compute spectral centroid (power-weighted mean frequency), bandwidth (frequency spread), and peak frequency. Descriptor medians are taken within recording and then within patient; HF uses recording-date groups. Principal component analysis (PCA) of log-transformed, jointly standardized descriptors with equal source weights shows source structure with partial overlap (Figure 1(b)).

To examine whether differences persist within the same

sound categories, we compare ICBHI and SPRSound within Normal, Crackle, Wheeze, and Both. SPRSound’s Fine/Coarse Crackle labels are combined as Crackle, and Wheeze+Crackle corresponds to Both. ICBHI has lower spectral centroids in every class (Figure 1(c)), showing that input differences persist after aligning the sound labels. Joint learning must therefore use compatible acoustic supervision across differing source distributions.

3. METHOD

LSAA connects heterogeneous annotations through shared abnormality (A), Crackle (C), and Wheeze (W) probabilities and dataset-specific readouts (Figure 2).

3.1. Shared Attributes and Task Readouts

The prediction path maps each audio unit to shared attribute probabilities and then to the output required by its dataset. For a mono 16-kHz, 5-s input x_i prepared as in Section 2.2, AudioSet-pretrained BEATs iter3+ AS2M [11, 12] produces $H_i \in \mathbb{R}^{T_i \times D}$, where T_i is the token count and $D = 768$. Mean pooling over tokens followed by a linear $768 \rightarrow 256$ projection gives $h_i \in \mathbb{R}^{256}$. Three parallel linear heads predict

$$p_{ik} = \phi_k(W_k h_i + b_k), \quad k \in \{a, c, w\}, \quad (1)$$

where ϕ_a returns the Abnormal probability from a two-logit softmax, and ϕ_c, ϕ_w apply sigmoid to one logit each. Their outputs form the shared attribute vector $\mathbf{p}_i = (p_{ia}, p_{ic}, p_{iw})$.

Let $\hat{a}_i$ be the Normal/Abnormal argmax (ties select Normal), $\hat{c}_i = \mathbb{I}[p_{ic} \geq \tau_c]$, and $\hat{w}_i = \mathbb{I}[p_{iw} \geq \tau_w]$. SPRSound uses $\hat{a}_i$ for Normal/Adventitious classification. For ICBHI,

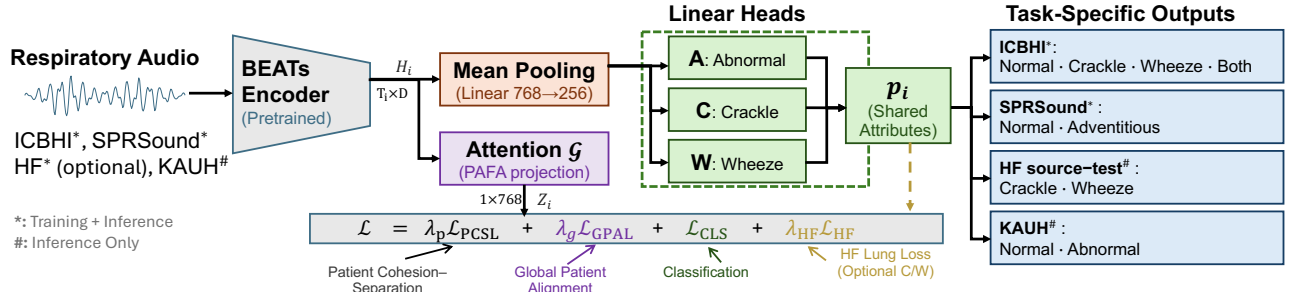


Fig. 2. LSAA prediction and training paths. BEATs tokens of width $D = 768$ are mean-pooled and linearly projected to 256 dimensions. Parallel heads form shared abnormality (A), Crackle (C), and Wheeze (W) probabilities. Task readouts convert these probabilities to dataset-specific outputs. A separate attention-based projection $\mathcal{G}$ supplies 768-dimensional features for the PAFA patient regularizers. ICBHI and SPRSound provide core attribute supervision and patient groups; the optional HF Lung term uses only source-train C/W supervision. HF source-test and KAUH evaluate fixed models and thresholds. The objective and readouts are defined in Section 3.

$\hat{a}_i = 0$ yields Normal; otherwise, two positive attributes yield Both and one yields its class. If neither is positive, the larger $p_{ik} - \tau_k$ margin selects Crackle or Wheeze, with ties selecting Crackle. Thresholds use core validation.

For HF source-test, maxima over C/W window probabilities give recording scores. For KAUH, we average probabilities across each patient’s filter versions and predict Normal/Abnormal for patients with compatible sound labels. Models and thresholds remain fixed for these evaluations.

3.2. Annotation-Aware Joint Training

Training uses each annotation to supervise the attributes it supports, jointly fine-tuning the encoder and prediction heads on ICBHI and SPRSound. Annotations supply targets y_{ik} and masks m_{ik} , with $m_{ik} = 1$ only for supported attributes, distinguishing unknown labels from observed negatives [13, 14]. ICBHI labels specify A/C/W. SPRSound’s fine C/W labels supervise the corresponding heads despite its binary task, while Rhonchi/Stridor supervise A only. For batch B , let $B_k = \{i \in B : m_{ik} = 1\}$ and $K_B = \{k : |B_k| > 0\}$. Classification uses

$$\mathcal{L}_{\text{cls}}(B) = \frac{1}{|K_B|} \sum_{k \in K_B} \frac{1}{|B_k|} \sum_{i \in B_k} \ell_k(p_{ik}, y_{ik}), \quad (2)$$

where ℓ_a is cross-entropy on $(1 - p_{ia}, p_{ia})$ and ℓ_c, ℓ_w are binary cross-entropy.

Patient-based regularization uses a separate attention-based projection $Z_i = \mathcal{G}(H_i) \in \mathbb{R}^{768}$ and PAFA’s patient cohesion-separation (PCSL) and global patient alignment (GPAL) losses [15]. Within each core-source batch, PCSL promotes within-patient cohesion and between-patient separation, while GPAL limits the dispersion of patient feature means. These losses regularize the shared encoder during joint training.

In the auxiliary condition, HF source-training windows add C/W supervision, mapping the discontinuous adventitious-sound label D to C. Their masked binary cross-entropy gives

$\mathcal{L}_{\text{HF}}$. The treatment of derived negatives is discussed in Section 4.2. Core batches supply classification and PAFA regularization, yielding

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{cls}} + \lambda_p \mathcal{L}_{\text{PCSL}} + \lambda_g \mathcal{L}_{\text{GPAL}}}_{\mathcal{L}_{\text{core}}} + \lambda_{\text{HF}} \mathcal{L}_{\text{HF}}, \quad (3)$$

with $\lambda_p = 50$, $\lambda_g = 5 \times 10^{-4}$, and $\lambda_{\text{HF}} = 0$ for core training or 0.25 for the auxiliary condition. HF contributes only the C/W term, leaving A supervision and patient regularization to the core sources.

4. EVALUATION

4.1. Joint Learning and Transfer

Table 1 compares LSAA with separately trained frozen-encoder references. LSAA achieves higher means on ICBHI and both external targets, whereas the strongest frozen references perform better on SPRSound. The ICBHI-only and SPRSound-only controls in Table 2(a) assess joint training within LSAA’s shared architecture.

Table 1. Performance of LSAA and frozen-encoder references across four respiratory sound datasets. Values are means (%) over three seeds, with sample standard deviations shown as subscripts. In Train, I/S denotes separate ICBHI/SPRSound models and I+S denotes joint training. **Bold** and underlined values mark the highest and second-highest means within each metric column. For LSAA, CAS AUROC evaluates maximum-window Wheeze-head scores against the Wheeze/Rhonchi/Stridor label union.

Model	Train	ICBHI	SPRSound	HF	KAUH
		[3]	[4]	[1]	[9]
		Score	Score	CAS AUROC	BA
AST [16]	I/S	51.40 $\pm$ 0.48	90.20 $\pm$ 0.18	<u>77.01</u> $\pm$ 0.30	<u>68.64</u> $\pm$ 0.41
BEATs [11]	I/S	51.68 $\pm$ 1.19	92.37 $\pm$ 0.35	75.87 $\pm$ 0.11	61.86 $\pm$ 0.34
PANNs [17]	I/S	<u>52.08</u> $\pm$ 0.16	<u>91.23</u> $\pm$ 0.55	73.13 $\pm$ 0.10	56.52 $\pm$ 2.24
OPERA-CT [2]	I/S	39.96 $\pm$ 1.68	79.45 $\pm$ 0.00	60.96 $\pm$ 0.35	59.88 $\pm$ 0.34
HeAR [18]	I/S	49.50 $\pm$ 0.62	87.89 $\pm$ 0.31	64.20 $\pm$ 0.29	64.10 $\pm$ 0.11
LSAA	I+S	61.17 $\pm$ 0.31	90.70 $\pm$ 0.34	86.44 $\pm$ 2.13	72.17 $\pm$ 1.64

Providing both readouts does not make single-source training sufficient for both datasets: each single-source model performs poorly on the other native task. These failures accompany the within-label acoustic differences in Figure 1(b,c). Joint supervision enables one model to serve both tasks, improving ICBHI Score by 3.39 percentage points (pp) at a 2.13-pp cost on SPRSound relative to their respective single-source models. Both Normal recall and correct abnormal-subclass recall improve on ICBHI. Joint training therefore expands task coverage, with different gains and costs relative to the two specialist models.

External benefits depend on the source and prediction target. LSAA’s HF CAS ranking AUROC (86.44%) is close to ICBHI-only (87.05%) and exceeds SPRSound-only (79.25%), while its KAUH patient BA exceeds both controls (Table 2(a)). Combining sources thus improves KAUH performance without producing a uniform gain across external targets.

4.2. Supervision and Readout Analysis

A binary prediction target can still benefit from fine supervision. Retaining SPRSound C/W labels improves both native Scores and SPRSound C/W AUROC across three seeds (Table 2(b)). Coarse SPR’s mean ICBHI Score falls below ICBHI-only, showing that adding source audio alone does not ensure a gain under this recipe. The SPRSound improvement is uneven: Coarse includes an all-Normal seed, while the other two seeds gain 4.57/0.15 pp. Coarse also removes C/W calibration and changes relative loss strength, so the comparison supports the fine-supervision recipe without isolating annotation information alone.

Table 2. Effects of training sources, supervision detail, and task readouts. (a) Absolute performance (%) and paired changes (pp), with Δ_I and Δ_S denoting LSAA minus ICBHI-only and SPRSound-only, respectively. (b) Variant minus LSAA; CW is SPRSound C/W macro AUROC (LSAA: 97.32 $\pm$ 0.15%). (c) HF-on minus the corresponding same-seed HF-off run. Values are means with subscripted $\pm$ sample SD over three seeds. IC/SPR: native Scores; CAS: HF recording AUROC using Wheeze-head scores for CAS-union ranking; BA: KAUH patient balanced accuracy.

(a) Single-source and joint training				
Model	IC	SPR	CAS	BA
ICBHI-only	57.78 $\pm$ 1.47	49.53 $\pm$ 6.40	87.05 $\pm$ 0.86	70.44 $\pm$ 1.52
SPRSound-only	47.75 $\pm$ 0.51	92.82 $\pm$ 0.65	79.25 $\pm$ 1.08	67.25 $\pm$ 4.29
LSAA	61.17 $\pm$ 0.31	90.70 $\pm$ 0.34	86.44 $\pm$ 2.13	72.17 $\pm$ 1.64
Δ_I	+ 3.39 $\pm$ 1.59	+41.17 $\pm$ 6.70	-0.61 $\pm$ 1.32	+1.73 $\pm$ 0.81
Δ_S	+13.42 $\pm$ 0.82	- 2.13 $\pm$ 0.33	+7.19 $\pm$ 1.05	+4.91 $\pm$ 4.69
(b) Fine supervision and readout				
Variant	Δ IC	Δ SPR	Δ CW	
Coarse SPR	-6.51 $\pm$ 4.33	-23.46 $\pm$ 36.62	-8.48 $\pm$ 8.58	
Native + C/W	+2.50 $\pm$ 1.90	+0.16 $\pm$ 1.55	-0.05 $\pm$ 0.25	
(c) HF supervision and transfer				
Variant	Native Δ Score		HF	KAUH
	IC	SPR	Δ CAS	Δ BA
HF-on	-1.90 $\pm$ 2.89	+1.31 $\pm$ 1.43	-7.41 $\pm$ 2.73	+1.91 $\pm$ 7.03

Shared fine supervision does not prescribe a unique task readout. Native + C/W improves ICBHI Score by 2.50 pp while SPRSound C/W AUROC changes by only -0.05 pp. This supports separate assessment of attribute detection and native-task performance. LSAA supplies an explicit attribute-to-task mapping; learned native heads offer another implementation. Because the alternative retrains both encoder and heads, the gain cannot be attributed solely to replacing the decoder.

More supervision from a source need not improve its evaluation target. HF-on lowers CAS AUROC in all three seeds (-7.41 $\pm$ 2.73 pp), despite improving SPRSound in two seeds; KAUH changes direction across seeds (Table 2(c)). HF training assigns W negatives to eligible Rhonchi/Stridor windows without Wheeze annotations, whereas CAS evaluation treats any Wheeze/Rhonchi/Stridor annotation as positive and ranks recordings using maximum-window Wheeze-head scores. Of 661 CAS-positive test recordings, 256 have no Wheeze annotation. Supervision and evaluation thus target different sound categories within the same source. This mismatch may contribute to the decline, but the current comparison does not isolate its causal effect.

5. CONCLUSION

We presented LSAA, a joint learning framework that connects heterogeneous respiratory sound annotations through

shared acoustic attributes. Task-specific readouts allow one model to support cycle-level four-class and event-level binary prediction while retaining fine-grained supervision. Our experiments support retaining fine annotations under the evaluated training recipe and reveal task-dependent tradeoffs from joint training and readout design. Evaluation on HF Lung and KAUH extends this assessment beyond the training datasets, showing that additional supervision does not uniformly improve transfer.

6. REFERENCES

- [1] F.-S. Hsu *et al.*, “Benchmarking of eight recurrent neural network variants for breath phase and adventitious sound detection on a self-developed open-access lung sound database—HF_Lung_V1,” *PLOS ONE*, vol. 16, no. 7, pp. e0254134, 2021.
- [2] Y. Zhang *et al.*, “Towards open respiratory acoustic foundation models: Pretraining and benchmarking,” in *Advances in Neural Information Processing Systems*, 2024, vol. 37, pp. 27024–27055.
- [3] B. M. Rocha *et al.*, “An open access database for the evaluation of respiratory sound classification algorithms,” *Physiological Measurement*, vol. 40, no. 3, pp. 035001, 2019.
- [4] Q. Zhang *et al.*, “SPRSound: Open-source SJTU paediatric respiratory sound database,” *IEEE Transactions on Biomedical Circuits and Systems*, vol. 16, no. 5, pp. 867–881, 2022.
- [5] S. Ge, W. Zhang, S. Xie, B. Yan, and Z. Wang, “Lung-Mix: A mixup-based strategy for generalization in respiratory sound classification,” in *ICASSP 2025—2025 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [6] Y.-W. Chua and Y.-C. Cheng, “Towards enhanced classification of abnormal lung sound in multi-breath: A light weight multi-label and multi-head attention classification method,” arXiv:2407.10828, 2024.
- [7] S. G. Jeong and S.-E. Kim, “PC-MCL: Patient-consistent multi-cycle learning with multi-label bias correction for respiratory sound classification,” in *ICASSP 2026—2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 2665–2669.
- [8] S. Cornell *et al.*, “DCASE 2024 task 4: Sound event detection with heterogeneous data and missing labels,” arXiv:2406.08056, 2024.
- [9] M. Fraiwan, L. Fraiwan, B. Khassawneh, and A. Ibnián, “A dataset of lung sounds recorded from the chest wall using an electronic stethoscope,” *Data in Brief*, vol. 35, pp. 106913, 2021.
- [10] P. Welch, “The use of the fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms,” *IEEE Transactions on Audio and Electroacoustics*, vol. 15, pp. 70–73, 1967.
- [11] S. Chen *et al.*, “BEATs: Audio pre-training with acoustic tokenizers,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023, vol. 202 of *Proceedings of Machine Learning Research*, pp. 5178–5193, PMLR.
- [12] J. F. Gemmeke *et al.*, “Audio Set: An ontology and human-labeled dataset for audio events,” in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.
- [13] G. Shi, L. Xiao, Y. Chen, and S. K. Zhou, “Marginal loss and exclusion loss for partially supervised multi-organ segmentation,” *Medical Image Analysis*, vol. 70, pp. 101979, 2021.
- [14] D. Zhou, P. Chen, Q. Wang, G. Chen, and P.-A. Heng, “Acknowledging the unknown for multi-label learning with single positive labels,” in *Computer Vision—ECCV 2022*, 2022, vol. 13684 of *Lecture Notes in Computer Science*, pp. 423–440, Springer.
- [15] S. G. Jeong and S. E. Kim, “Patient-aware feature alignment for robust lung sound classification: Cohesion-separation and global alignment losses,” in *Interspeech 2025*, 2025, pp. 1018–1022.
- [16] Y. Gong, Y.-A. Chung, and J. Glass, “AST: Audio spectrogram transformer,” in *Interspeech 2021*, 2021, pp. 571–575.
- [17] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, “PANNs: Large-scale pretrained audio neural networks for audio pattern recognition,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2880–2894, 2020.
- [18] S. Baur *et al.*, “HeAR—health acoustic representations,” arXiv:2403.02522, 2024.